E5 – Chemistry modelling for an urban area with traffic emissions

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Description

This exercise is about using the chemistry model. The aim is to learn:

- how to switch on chemistry
- about the main chemistry options
- how to apply different chemistry mechanisms
- how to apply different options for anthropogenic emissions

Requirements

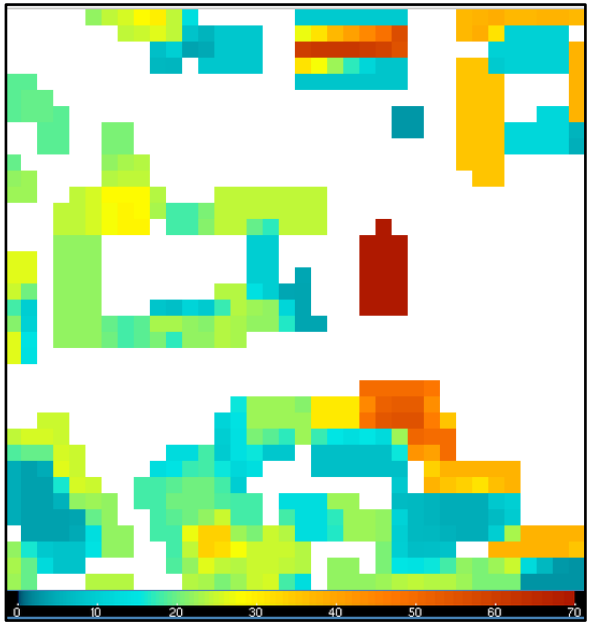
- Lecture „Chemistry model“
- Lecture „Static and dynamic drivers“

Chemistry model exercise

Summary of simulation setup

General setup

- Small part of Berlin with realistic distribution of streets and buildings
- 10 m grid width
- Cyclic boundary conditions
- Short runs for instationary conditions around sunrise
- **Baseline** run for **photostationary equilibrium** between NO, NO_2 and O_3 plus one passive tracer and parameterized traffic emissions



Building heights from the static file for this exercise

Note that the quality of the results is limited due to the small size of the model domain and the coarse resolution (mainly a meteorology issue – the chemistry will run stable nevertheless).

Input files to start with the excercise are supplied in:
E-5_chemistry_steering_files_incomplete.zip

General setup

Input files to start with the excercise are supplied in
=> **E-5_chemistry_steering_files_incomplete.zip**

Containing files:

- **`chem_phstatp_p3d`** (p3d with simple chemistry namelist for phstatp mechanism)
- **`chem_phstatp_static`** (static file for the area shown on the slide before)
- **`chem_phstatp_lod2_chemistry`** (time dependent traffic emissions for Task 3)

For completeness, the directory structure including the ready-to run files for the exercise are supplied in E-5_chemistry_steering_files_complete.zip.

Please do not look into E-5_chemistry_steering_files_complete.zip before you tried to solve the exercise yourself!

Task 1: Simulation with the phstatp mechanism from 06 UTC to 08 UTC (baseline case)

- **Simulation time**: 06 to 08 UTC
- **Mechanism:** photostationary equilibrium between NO, NO_2 and O_3 plus one passive tracer named PM10 (mechanism **phstatp**)
- **Emissions:** parameterized traffic emissions (`emiss_lod = 0`)

Create and run a simulation setup named `chem_phstatp`

1. Create a simulation setup `chem_phstatp` in the `JOBS` directory.
   Hint: Use the input files `chem_phstatp_p3d` and `chem_phstatp_static` supplied in `E-5_chemistry_steering_files.zip`
2. Run your setup for `chem_phstatp`
3. Look into the results

Preliminary question

(1) Where can I expect high traffic emissions in my domain?
Hint: What is the meaning of `emiss_lod = 0`?
Look into the static file

Question on simulation results

(View results with ncview or a similar simple viewer - with ncview, you can get time series by clicking at a point)

How do the concentrations of PM10, NO, NO_2 and ozone correlate?
Look at area plots at the end of the simulation and at time series for ozone, NO and NO2, and PM10 at x=285m and y=185m.

Task 2: Simulation with the smog mechanism from 06 UTC to 08 UTC

Create and run a simulation setup with run_identifier named `chem_smog`

1. Use **`chem_phstatp_p3d`** as a basis for your namelist file **`chem_smog_p3d`** and use the same static file as in Task 1
   Hint: Preferrably use a soft link to obtain **`chem_smog_static`** instead of copying the static file into **`chem_smog`**

2. Create a **`USER_CODE`** directory in **`JOBS/chem_smog`** and add **`chem_gasphase_mod.f90`** for smog into this directory
   Hint: Look into **`packages/chemistry/kpp4palm/mechanisms`**

Create and run a simulation setup named `chem_smog` (cont.)

3. Edit your **`chem_smog_p3d`**

   a. Include parameterized surface emissions (**`surface_csflux`**) to the namelist for RH (649 μmol m^{-2} d^{-1}), RCHO (30 μmol m^{-2} d^{-1}), and CO (30000 μmol m^{-2} d^{-1})
   Hint1: smog does not conclude PM10
   Hint2: Do not forget to extend the number of factors for **`emiss_factor_main/side`** by the number of new variables.

   b. Additional initial values/profiles:
   RH: linear decrease of 0.010 ppm at surface to 0.005 ppm at 295 m, constant above
   RCHO: same profile as for RH, but only half of the concentration of RH
   CO: 0.1 ppm at all heights

   c. Add instantaneous 3d output for the three new initialized variables (RH, RCHO, CO) and for concentrations of OH, HO2 and HNO3

Create and run a simulation setup named `chem_smog` (cont.)

4. Start a simulation with your setup chem_smog
   (Will take much longer than for phstatp. If the simulation takes too long, re-run the simulation for a shorter time)

5. Analyze results e.g. with ncview.

   - Compare and explain the differences in the concentrations of ozone for phstatp and smog at 5 m above the ground (at 7200 sec if available or for an earlier time).

   - Look at time series for ozone, NO2 and NO, products and radicals at x=285m and y=185m at a height of 5 m.

Preliminary questions

(1) Where can we find the **`chem_gasphase_mod.f90`** for smog?

(2) What about PM10 when running PALM with smog? Keep or remove? How can we find out?
Hint: Check either **`chem_gasphase_mod.f90`** or the .eqn file in the respective subdirectory of **`packages/chemistry/kpp4palm/mechanisms`**

Question on simulation results

Why is ozone higher for smog than for phstatp?

Hint: Check either **`chem_gasphase_mod.f90`** or the .eqn file in the respective subdirectory of **`packages/chemistry/kpp4palm/mechanisms`**

Task 3: Run with LOD 2 emission input and phstatp mechanism

Create and run a simulation setup named `chem_phstatp_lod2`

- Create the simulation setup **`chem_ phstatp_lod2`** by adapting **`chem_phstatp_p3d`** concerning the emission input (LOD 2 instead of LOD 0)
- Remove namelist parameters that are only needed for the parameterized emissions of LOD 0 (**`surface_csflux*, *street*)`**
- Use the same static file as in Task 1 and Task 2 and copy the chemistry driver **`chem_phstatp_chemistry,`** which contains the pre-processed real-time traffic emissions to the INPUT directory (details on LOD 2 emissions are on next slide)
- Run the new setup and compare it to the results of **`chem_phstatp`**.
- Analyze the concentration results using ncview

Chemistry model exercise

Exercise Task 3

Short description of LOD 0 daily traffic emission values and LOD 2 traffic emission estimate

- **LOD 0**
  - Estimate using HBEFA 3.3 emission factors for exhaust emissions and mean traffic counts for Berlin differing main and side roads only. Temporal disaggregation based on a fixed week-day standard urban time profile
  - Main and side roads identified using the street_type classification from OpenStreetMap
  - For PM10 only emissions of car exhaust are considered. **No particle resuspension processes are taken into account!**
- **LOD 2**
  - Data based on calculations applying an emission model
  - Using HBEFA 3.3 emission factors (INFRAS, 2017) and hourly real-time traffic counts, temperature time series and the most current car fleet estimate for Berlin
  - **PM10 emissions include also emissions from particle resuspension** by using a corresponding parameterization for this emission source

Questions Task 3

Preliminary question

Take a look at the emission fields within the chemistry driver `chem_phstatp_lod2_chemistry` using e.g. ncview.
What is eye-catching when looking into the data?

Hint: Change "Axes" and adapt the „Range" in ncview to see the emissions of the different compounds (i.e. NO, NO_2 and PM10). Check „`emission_name`" in the nc file using „ncdump" to identify the different compounds (`ncdump -v emission_name chem_phstatp_lod2_chemistry`)

Question on simulation results

How much do the results using LOD 2 emission input differ from those of Task 1? Can you explain the differences in PM10 based on the description of the different emission inputs?

HINT: See previous slide and the description for the different LODs on https://palm.muk.uni-hannover.de/trac/wiki/doc/app/chememi or in the PALM model system on GitLab under:
palm_model_system-v23.04\packages\palm\model\docs\content\Guide\Modules\Chemistry